ModelIt!

Student Activity Pack

CRISPR Gene Drive: Should We Eliminate Malaria?

Systems Biology Modeling of Gene Drive Ecology and Ethics

Grade 09L3 | NGSS HS-LS3-2, HS-LS4-5 | Cause and Effect / Systems and System Models

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you know about CRISPR and how it is used to edit genes?

2 Malaria kills about 600,000 people per year — mostly children. If we could eliminate the disease by making mosquitoes extinct, should we?

3 Draw a food web that includes mosquitoes — what other species would be affected if mosquitoes disappeared?

4 What does it mean for a gene to 'drive' through a population, and how is this different from normal inheritance?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

- Gene Drive Efficiency
- Mosquito Population
- Malaria Transmission Rate
- Wild-Type Gene Frequency
- Ecological Dependence
- Resistance Evolution
- Geographic Spread Rate
- Human Malaria Cases
- Ecosystem Disruption Risk

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Moderate Regional Deployment

95% efficiency, single region release

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Resistance Race

85% efficiency, elevated resistance evolution

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Global Maximum Release

99.5% efficiency, global spread

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Alternative Vector Species — Other mosquito species or biting insects that could potentially fill the ecological niche and become new malaria vectors if Anopheles mosquitoes are suppressed

Gene Drive Reversal Mechanism — An engineered 'daisy chain' or 'kill switch' system designed to halt or reverse gene drive spread — still theoretical and unproven in wild populations

Climate Change Interaction — The effect of changing temperature and precipitation patterns on mosquito range, breeding habitats, and gene drive spread dynamics — climate change is expanding malaria zones regardless of gene drive deployment

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

We have the technology to make mosquitoes extinct and end malaria forever — but should we?
Use evidence from your simulation to support your answers.

1. **How does your systems biology model reveal the tension between saving human lives and protecting ecosystem stability?**

2. **Why is it important that gene drives cannot be geographically contained — what makes this different from other genetic engineering applications?**

3. **What role does Resistance Evolution play in determining whether the gene drive succeeds or fails, and how predictable is this outcome?**

4. **How would adding a Gene Drive Reversal Mechanism change your model's predictions and the governance conversation?**

5. **What are the ethical implications of a technology that makes irreversible changes to wild species — who has the right to make that decision?**

8 STEM Challenge: Design a Gene Drive Governance Framework

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YOUR ENGINEERING MISSION

Design a comprehensive governance framework that determines when, where, and how gene drive technology should be deployed — balancing human health benefits against ecological risks using systems biology evidence.

THE SCENARIO

The United Nations has convened a special panel on gene drive technology. Malaria kills 600,000 people per year, mostly children under 5 in sub-Saharan Africa. Gene drive technology could potentially eliminate the disease. But the technology is irreversible once deployed and could have unpredictable ecological consequences. Your team has been appointed to design the governance framework that will determine global policy on gene drive deployment.

Engineering Design Thinking

1. HOW WOULD YOU DESIGN A 'KILL SWITCH' OR RECALL MECHANISM FOR THE GENE DRIVE IF UNINTENDED CONSEQUENCES ARE DETECTED?

2. WHAT INTERNATIONAL AGREEMENTS WOULD NEED TO EXIST BEFORE A GENE DRIVE THAT CROSSES NATIONAL BORDERS IS RELEASED?

3. HOW DO YOU WEIGH 600,000 HUMAN DEATHS PER YEAR AGAINST UNCERTAIN ECOLOGICAL RISKS THAT MAY OR MAY NOT MATERIALIZE?

4. WHAT ROLE SHOULD THE COMMUNITIES MOST AFFECTED BY MALARIA PLAY IN THE DECISION-MAKING PROCESS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Gene Drive Researchers and Conservation Geneticists work at the intersection of genetic engineering and ecology. They develop and evaluate gene drive technologies at institutions like the Target Malaria consortium, DARPA, and academic research labs, earning $80,000–$160,000/year. Bioethicists who evaluate the societal implications of these technologies earn $70,000–$140,000/year.